



ASCHAM SCHOOL

2019 YEAR 12
MATHEMATICS EXTENSION 2
TRIAL EXAM

GENERAL INSTRUCTIONS

Reading time – 5 minutes

Working time – 180 minutes

Use black pen, non-erasable

NESA-approved calculators may be used

Reference Sheet is provided

Total Marks - 40

Section A – Multiple Choice (1 mark each)

Attempt Questions 1 to 10.

Select answers on the separate multiple choice sheet provided.

Write your NESA number on the multiple choice sheet.

Section B – Questions 11 – 16 (15 marks each)

Start each question in a new booklet.

If you use a second booklet for a question, place it inside the first.

Label extra booklets for the same question as, for example, Q11-2 etc.

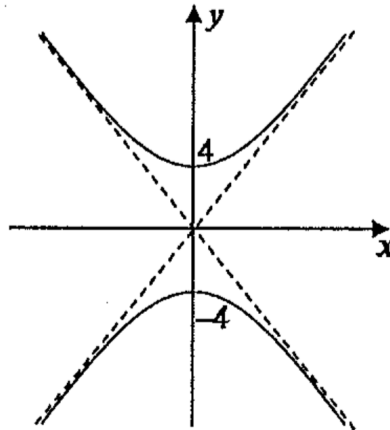
Write your NESA number and question number on each booklet.

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Section A - Multiple choice (10 marks)

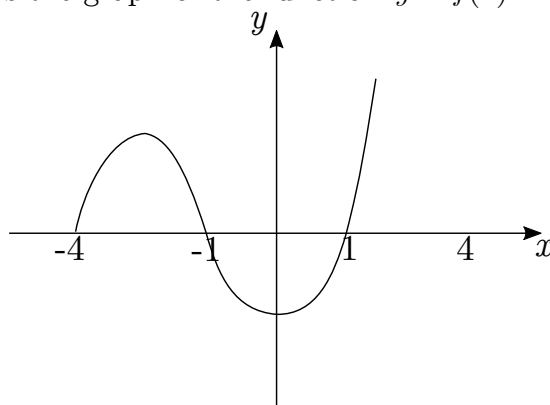
(Mark the correct answer on the sheet provided.)

1. Which is a possible equation of the following hyperbola?



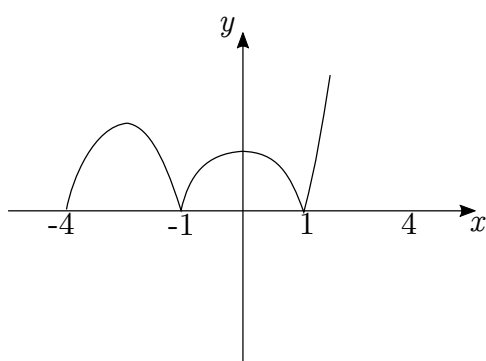
- A) $9x^2 - 16y^2 = 144$ B) $16x^2 - 9y^2 = 144$
- C) $9x^2 - 16y^2 = -144$ D) $16x^2 - 9y^2 = -144$
2. $\frac{2-i}{-2-i} = ?$
- A) $-\frac{3}{5} + \frac{4}{5}i$ B) -1
- C) $-1 + \frac{4}{3}i$ D) $-\frac{5}{3}$

3. The diagram shows the graph of the function $y = f(x)$.

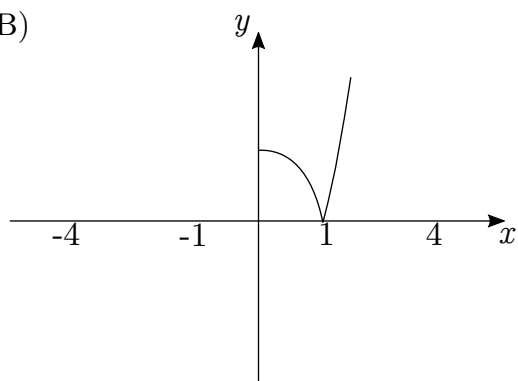


Which of the following is the graph of $y = f(|x|)$?

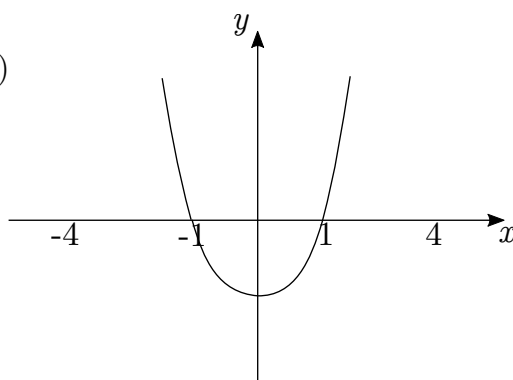
A)



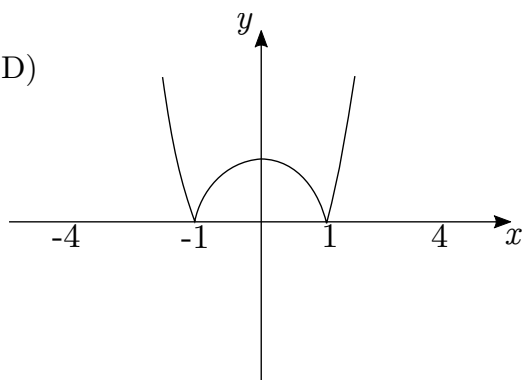
B)



C)



D)



4. The polynomial equation $x^3 + Ax^2 + B = 0$ has roots α , β and γ . What are the roots of the polynomial equation $(3x + 2)^3 + A(3x + 2)^2 + B = 0$?

A) $\frac{\alpha}{3} - 2, \frac{\beta}{3} - 2, \frac{\gamma}{3} - 2$

B) $\frac{\alpha - 2}{3}, \frac{\beta - 2}{3}, \frac{\gamma - 2}{3}$

C) $3\alpha + 2, 3\beta + 2, 3\gamma + 2$

D) $\alpha + \frac{2}{3}, \beta + \frac{2}{3}, \gamma + \frac{2}{3}$

5. Given the rectangular hyperbola $xy = 25$, which is the correct equation of its directrices?

A) $x + y = \pm \frac{5}{\sqrt{2}}$

B) $x + y = \pm 5$

C) $x + y = \pm 5\sqrt{2}$

D) $x + y = \pm 10$

6. Let α, β and γ be the zeroes of the polynomial $x^3 + 5x - 3$. Find the value of $\alpha^3 + \beta^3 + \gamma^3$.

A) -125

B) 0

C) 9

D) 34

7. The equation $|z - 3| + |z + 3| = 10$ defines an ellipse. What is the length of the semi-minor axis?

A) 4

B) 5

C) 8

D) 10

8. The complex number z satisfies the equation $|z - 2| = 1$. What is the maximum value of $\arg(z)$?

A) $\tan^{-1}\left(\frac{1}{2}\right)$

B) $\tan^{-1}\left(\frac{1}{\sqrt{3}}\right)$

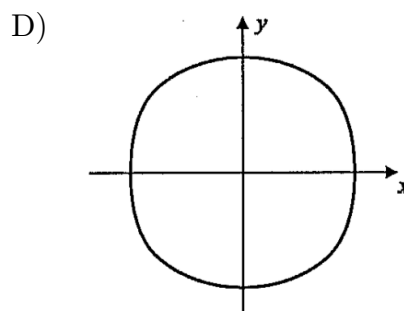
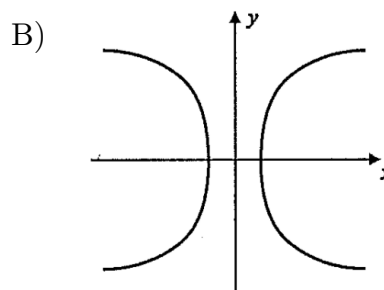
C) $\tan^{-1}1$

D) $\tan^{-1}\sqrt{3}$



10. After differentiating a relation implicitly, we find that $\frac{dy}{dx} = \frac{-\sqrt[3]{y}}{\sqrt[3]{x}}$.

Which of the following could be a graph of this relation?



6

Section B (35 marks)

Question 11 (Begin and label a new booklet.)

(15 marks)

a) Evaluate $\int_1^e \log_e x \, dx$. [2]

b) Evaluate $\int_0^\pi \sin^3 x \, dx$. [3]

c) i) Find values A , B and C so that $\frac{x^2 + x + 1}{x^3 + 3x^2} = \frac{A}{x} + \frac{B}{x^2} + \frac{C}{x + 3}$. [3]

ii) Hence find $\int \frac{x^2 + x + 1}{x^3 + 3x^2} \, dx$. [1]

d) Find $\int \frac{dx}{2 + \sin 2x}$ using the substitution $t = \tan x$. Leave your answer in terms of t . [3]

e) The polynomial $P(x) = x^5 + 2x^4 + ax^3 + bx^2$ has $(x - 1)^2$ as a factor.

Show that $a = -7$ and $b = 4$. [3]

(End of Question 11.)

Question 12 (Begin and label a new booklet.)

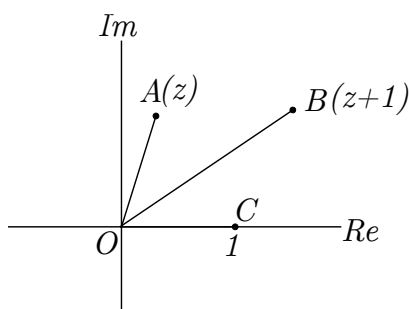
(15 marks)

a) Solve $z^2 = 5 - 12i$, giving your answer/s in the form $x + iy$. [2]

b) i) Express $-1 - i$ in modulus argument form. [2]

ii) Hence find the real part of $(-1 - i)^{10}$. [2]

c) As shown, the points O, C, A and B on the Argand diagram represent the complex numbers $0, 1, z$ and $z + 1$ respectively, where $z = \cos \theta + i \sin \theta$, $0 < \theta < \pi$. Copy the diagram.



i) Explain why $OCBA$ is a rhombus. [1]

ii) Draw the vector $z - 1$ on the diagram and hence explain why $\frac{z - 1}{z + 1}$ is purely imaginary. [2]

iii) Find, in terms of θ , the modulus and argument of $z + 1$. [2]

d) Let ω be a non-real root of $z^7 - 1 = 0$.

i) Show that $1 + \omega + \omega^2 + \omega^3 + \omega^4 + \omega^5 + \omega^6 = 0$. [1]

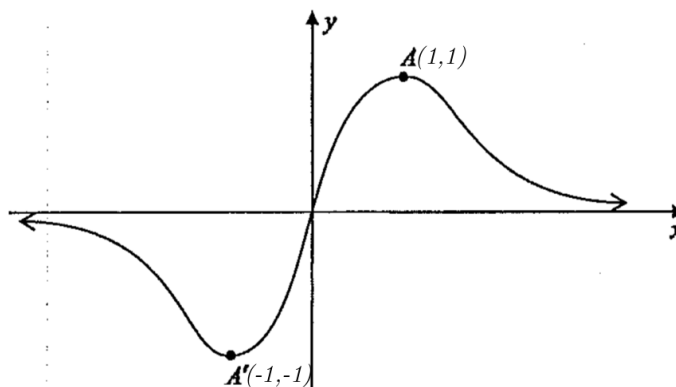
ii) Simplify $(\omega + \omega^2 + \omega^4)(\omega^6 + \omega^5 + \omega^3)$. [2]

iii) Sketch and label on the Argand diagram all seven roots of $z^7 - 1 = 0$. (You are not required to derive them.) [1]

(End of Question 12.)

Question 13 (Begin and label a new booklet.)**(15 marks)**

- a) Drawn below is the graph of $y = \frac{2x}{1+x^2}$. Stationary points at A and A' are labeled as shown.



Sketch on separate axes, labeling any important features:

i) $y = \frac{|2x|}{1+x^2}$ [1]

ii) $y = \frac{1+x^2}{2x}$ [2]

iii) $y^2 = \frac{2x}{1+x^2}$ [2]

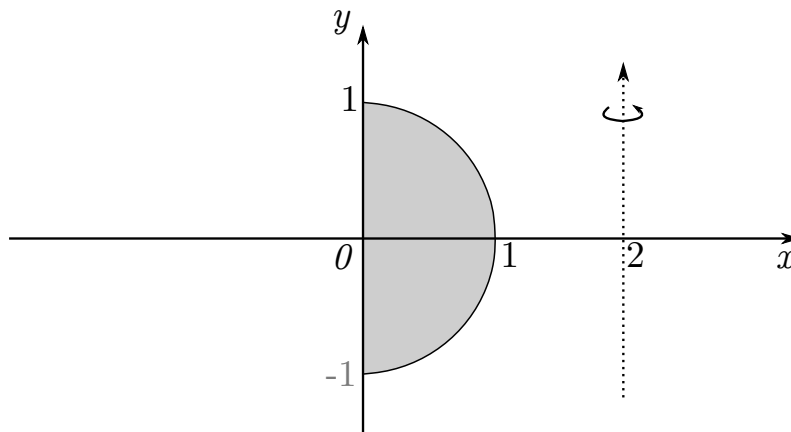
iv) $y = \log_e \left(\frac{2x}{1+x^2} \right)$ [2]

- b) Sketch the region on the Argand diagram that satisfies:

$$-\frac{2\pi}{3} \leq \arg(z-2) \leq 0 \text{ and } \operatorname{Im}(z) \leq -2\sqrt{3} \quad [2]$$

(Question 13 continues on the next page...)

- c) The shaded semicircle in the diagram below is rotated about the line $x = 2$.



- i) Using the method of cylindrical shells, show that the volume V of the resulting solid is given by:

$$V = \int_0^1 4\pi(2-x)\sqrt{1-x^2} \, dx. \quad [3]$$

- ii) Hence find the volume of the solid. [3]

(End of Question 13.)

Question 14 (Begin and label a new booklet.)

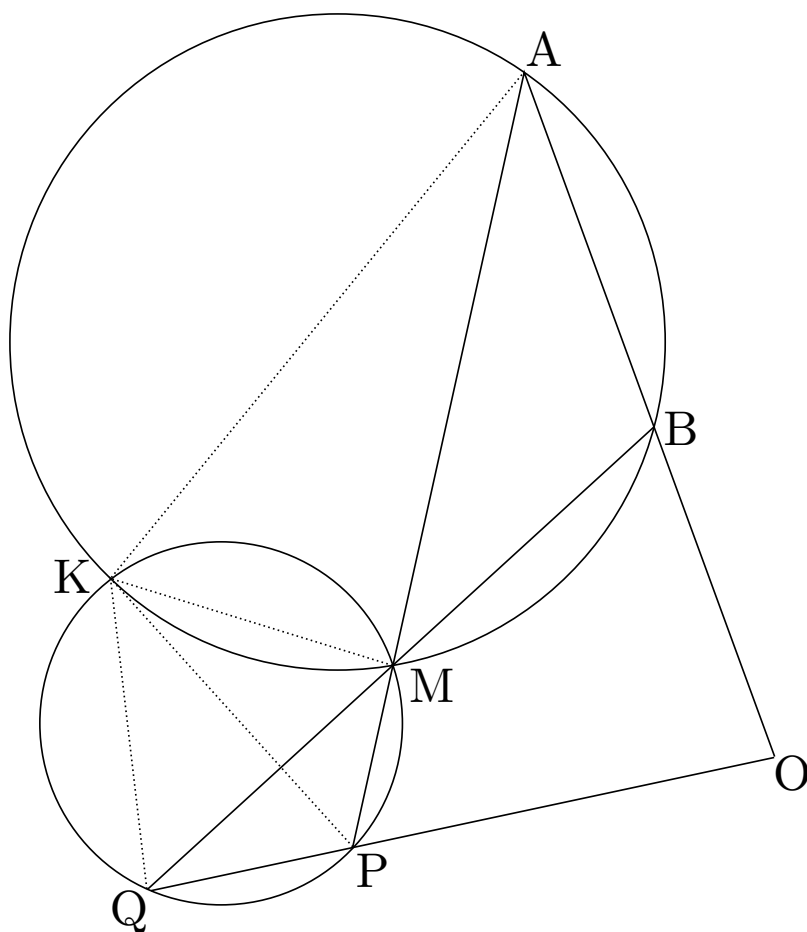
(15 marks)

- a) Suppose that x and y are positive. Prove that $\frac{1}{x} + \frac{1}{y} \geq \frac{4}{x+y}$. [3]
- b) An object of mass 5 kg is dropped in a medium where the resistance at speed v m/s has a magnitude of v Newtons. The acceleration due to gravity is 10 m/s^2 .
- i) Taking downwards as the positive direction, draw a force diagram and show that the equation of motion is $\ddot{x} = \frac{50-v}{5}$. [2]
- ii) Find an expression for the velocity v at time t seconds after the object is dropped. [2]
- iii) Find the terminal velocity of the object. [1]
- iv) Show that the distance x metered travelled when the speed is v m/s is given by:

$$x = 250 \log_e \left(\frac{50}{50-v} \right) - 5v \quad [2]$$

(Question 14 continues on the next page...)

- c) The diagram above shows two circles intersecting at K and M . From points A and B on the outer arc of one circle, lines are drawn through M to meet the other circle at P and Q respectively. The lines AB and QP meet at O .



- i) Let $\theta = \angle KAB$, and give a reason why $\angle KMQ = \theta$.
[1]
- ii) Prove $AKPO$ is a cyclic quadrilateral. [2]
- iii) Let $\phi = \angle AKM$. Show that if $OBMP$ is a cyclic quadrilateral, then the points A , K and Q are collinear. [2]

(End of Question 14.)

Question 15 (Begin and label a new booklet.)

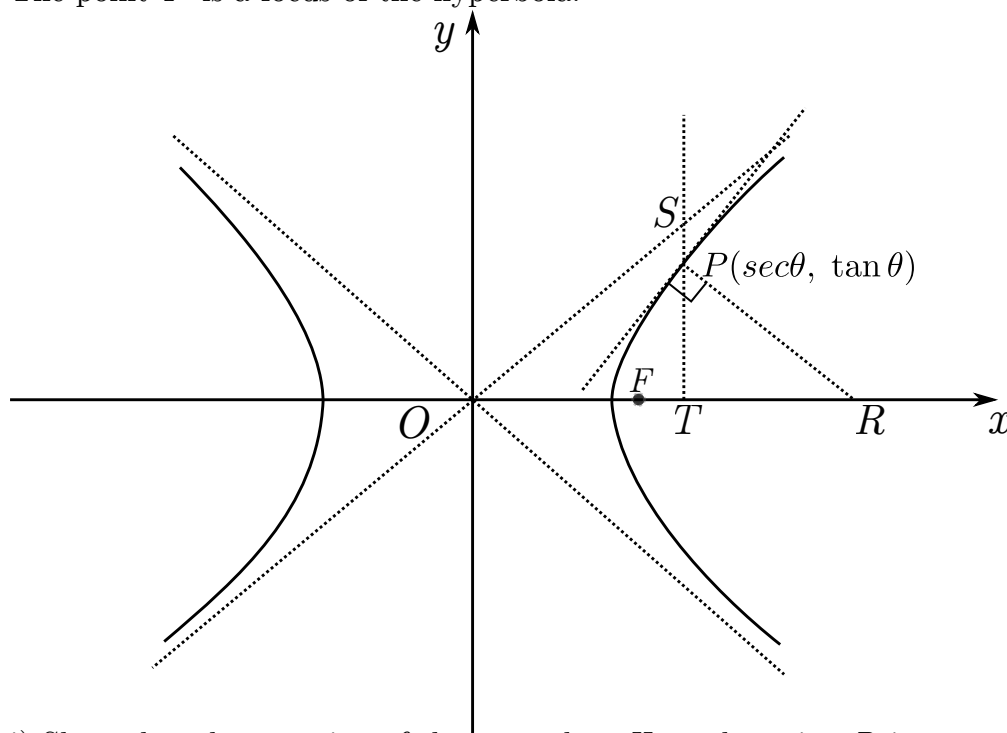
(15 marks)

a) Suppose that a, b and c are the side lengths of a triangle.

i) Explain why $(b - c)^2 < a^2$. [1]

ii) Deduce that $(a + b + c)^2 < 4(ab + bc + ca)$ [2]

b) The point $P(\sec \theta, \tan \theta)$ lie on the hyperbola with equation $x^2 - y^2 = 1$. A vertical line through P intersects with an asymptote at S and with the x -axis at T as shown. The normal to the hyperbola at P intersects the x -axis at R . The point F is a focus of the hyperbola.



i) Show that the equation of the normal to H at the point P is $y = -x \sin \theta + 2 \tan \theta$. [2]

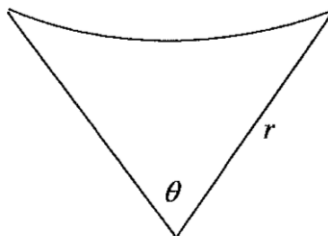
ii) Show that $RS = \sqrt{2} \times RT$. [2]

iii) Find the coordinates of the point W which lies on SR such that TW is parallel to the asymptote on which S lies. [2]

iv) For what values of θ will FW be the perpendicular bisector of SR ? [1]

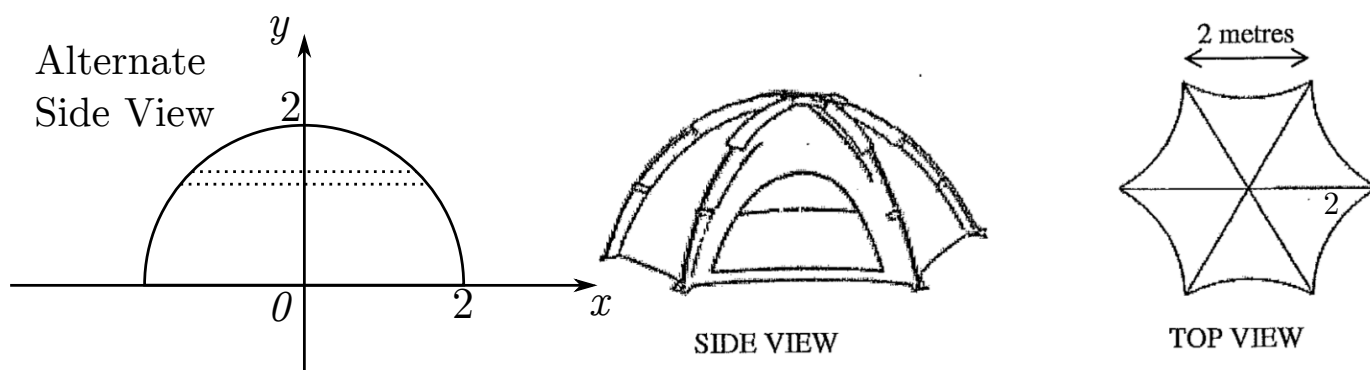
(Question 15 continues on the next page...)

- c) i) A sector is normally made up of a triangle and a minor segment of a circle. The shape below is made by subtracting the area of the minor segment from the triangle.



Show that its area is given by $A = r^2 \left(\sin \theta - \frac{\theta}{2} \right)$. [1]

- ii) A dome tent is built with a base made up of six congruent copies of the shape from (i), each with radius 2 metres. The tent is supported by flexible exterior poles extended between opposite corners in semi-circular arcs.



- iii) By taking slices parallel to the base of the tent, show that the volume enclosed by the tent is $\left(16\sqrt{3} - \frac{16\pi}{3} \right)$ cubic metres. [4]

(End of Question 15.)

Question 16 (Begin and label a new booklet.)**(15 marks)**a) Consider the integral $I_n = \int_0^1 x^{2n+1} e^{-x^2} dx$.i) Use integration by parts to show that $I_n = -\frac{1}{2e} + nI_{n-1}$, for $n \geq 1$. [3]ii) Show that $I_0 = \frac{1}{2} - \frac{1}{2e}$. [1]iii) Prove by mathematical induction that for all $n \geq 1$:

$$1 + \frac{1}{1!} + \frac{1}{2!} + \frac{1}{3!} + \dots + \frac{1}{n!} = e - \frac{2eI_n}{n!}. \quad [4]$$

iv) By considering the value of $x^{2n+1}e^{-x^2}$ in the domain $0 \leq x \leq 1$, explain why:

$$1 + \frac{1}{1!} + \frac{1}{2!} + \frac{1}{3!} + \dots = e \quad [1]$$

b) The numbers x, y and z satisfy:

$$x + y + z = 5$$

$$x^2 + y^2 + z^2 = 8$$

$$x^3 + y^3 + z^3 = 13$$

i) Show that $xy + xz + yz = \frac{17}{2}$. [1]ii) Show that $x^2y + x^2z + xy^2 + xz^2 + y^2z + yz^2 = 27$. [2]iii) Hence show that $xyz = \frac{31}{6}$. [1]iv) Use the previous parts to evaluate $x^4 + y^4 + z^4$. [2]*(End of Question 16.)**End of exam.*

Extension 2 2019 Trial Solutions

Section A

Q1. $x=0, y=\pm 4$

$$\therefore 16x^2 - 9y^2 = -144$$

$$\frac{y^2}{16} - \frac{x^2}{9} = 1$$

(D)

Q2. $\frac{2-i}{-2-i} = \frac{(2-i)(-2+i)}{4+1}$

$$= \frac{-4+2i+2i+1}{5}$$

$$= -\frac{3}{5} + \frac{4i}{5} \quad (A)$$

Q3. C $y=f(4x)$

(C)

Q4. $\alpha^3 + A\alpha^2 + B = 0$

$\therefore$ If $x = \frac{\alpha-2}{3}$, $3x+2 = \alpha \dots$
Satisfies new equation

(B)

Q5. $y - \frac{5}{\sqrt{2}} = -(x - \frac{5}{\sqrt{2}})$

$$x+y = \frac{5}{\sqrt{2}} \times 2$$

$$x+y = \sqrt{2} \times 5$$

(C)

Q6. $\alpha^3 + 5\alpha - 3 = 0$

$$\beta^3 + 5\beta - 3 = 0$$

$$\gamma^3 + 5\gamma - 3 = 0$$

$$\alpha^3 + \beta^3 + \gamma^3 + 5(\alpha + \beta + \gamma) - 9 = 0$$

$$\alpha + \beta + \gamma = 0$$

$$\therefore \alpha^3 + \beta^3 + \gamma^3 = 9$$

(C)

Q7.



$$a=5 \quad b^2 = a^2(1-e^2)$$

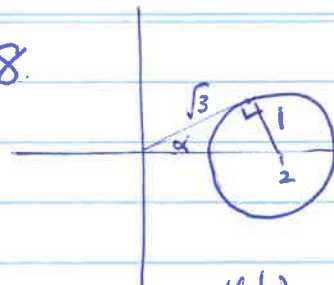
$$ae=3 \quad = 25(1-\frac{9}{25})$$

$$e=\frac{3}{5} \quad = 16$$

$$b=4$$

(A)

Q8.



$$\therefore \alpha = \tan^{-1}(\frac{1}{\sqrt{3}})$$

(B)

Q9. B or D since
root of mult. 3
at $x=2$.

Try: $y = x(x-2)^3$

$$y' = (x-2)^3 + x \cdot 3(x-2)^2$$

$$= (x-2)^2(x-2+3x)$$

$$= (x-2)^2(4x-2) \checkmark$$

(B)

(The other will not give
suitable y')

Q10. at $x \geq 0, y > 0$,

$$\frac{dy}{dx} < 0, \therefore \text{NOT A, B}$$

C+D: at $x=0, y \neq 0, \frac{dy}{dx} \rightarrow -\infty$
(in 1st quad)

vertical at $x=0$

(C)

1 2 3 4 5 6 7 8 9 10
DACBC CABBC

Section B

Q11 a) $\int_1^e \ln x \, dx$
 $= [x \ln x]_1^e - \int_1^e x \cdot \frac{1}{x} \, dx$
 $= (e \ln e - 1 \cdot \ln 1) - \int_1^e 1 \, dx$
 $= e - [x]_1^e$
 $= e - (e - 1)$
 $= 1$

b) $\int_0^\pi \sin^3 x \, dx$
 $= \int_0^\pi -\sin x (1 - \cos^2 x) \, dx \quad u = \cos x$
 $\quad \quad \quad du = -\sin x \, dx$
 $= -\int_1^{-1} 1 - u^2 \, du \quad x=0, u=1$
 $\quad \quad \quad x=\pi, u=-1$
 $= \int_{-1}^1 1 - u^2 \, du$
 $= [u - \frac{1}{3}u^3]_{-1}^1$
 $= (1 - \frac{1}{3}(1)) - (-1 - \frac{1}{3}(-1))$
 $= \frac{2}{3} - (-\frac{2}{3})$
 $= \frac{4}{3}$

c) i) $A(x+3)x + B(x+3) + Cx^2 = x^2 + x + 1$
 $x = -3: 9C = 9 - 3 + 1$
 $9C = 7$
 $C = \frac{7}{9} \checkmark$
 $x = 0: 3B = 1$
 $B = \frac{1}{3} \checkmark$

~~Let~~ Coeff x^2 :

$A + C = 1$
 $A = 1 - C = \frac{2}{9} \checkmark$

ii) $\int \frac{2}{9} \cdot \frac{1}{x} + \frac{1}{3} \cdot \frac{1}{x^2} + \frac{7}{9} \cdot \frac{1}{x+3} \, dx$
 $= \frac{2}{9} \ln x - \frac{1}{3} \cdot \frac{1}{x} + \frac{7}{9} \ln(x+3) + C$

d) $\int \frac{dx}{2 + \sin 2x} \quad t = \tan x \rightarrow x = \tan^{-1} t$
 $\sin 2x = \frac{2t}{1+t^2}$
 $dx = \frac{dt}{1+t^2}$
 $= \int \frac{1}{(2 + \frac{2t}{1+t^2})} \cdot \frac{dt}{1+t^2}$
 $= \int \frac{dt}{2 + 2t^2 + 2t}$
 $= \frac{1}{2} \int \frac{dt}{t^2 + t + 1}$
 $= \frac{1}{2} \int \frac{dt}{(t + \frac{1}{2})^2 + \frac{3}{4}}$
 $= \frac{1}{2} \cdot \frac{1}{\frac{\sqrt{3}}{2}} \tan^{-1} \left(\frac{t + \frac{1}{2}}{\frac{\sqrt{3}}{2}} \right) + C$
 $= \frac{1}{\sqrt{3}} \tan^{-1} \left(\frac{2t+1}{\sqrt{3}} \right) + C$

e) $P(1) = 0$
 $1 + 2 + a + b = 0$
 $a + b = -3 \quad \text{--- (1)}$
 $P'(1) = 0$

$P'(x) = 5x^4 + 8x^3 + 3ax^2 + 2bx$
 $\therefore 5 + 8 + 3a + 2b = 0$
 $3a + 2b = -13 \quad \text{--- (2)}$

$(2) - 2 \times (1): a = -13 - (-6)$
 $= -7$
 $\therefore -7 + b = -3$
 $b = 4$

Q12

a) $z^2 = 5 - 12i$

$(x+iy)^2 = 5 - 12i, x, y \in \mathbb{R}$

$x^2 - y^2 = 5$

$2xy = -12$

$\therefore y = \frac{-6}{x}$

$x^2 - \frac{36}{x^2} = 5$

$x^4 - 5x^2 - 36 = 0$

$(x^2 - 9)(x^2 + 4) = 0$

$\therefore x = \pm 3$

$x = 3, y = -2$

$x = -3, y = 2$

$\therefore z = 3 - 2i, -3 + 2i$

b) i) 

$|z| = \sqrt{1^2 + 1^2}$

$= \sqrt{2}$

$\arg z = \frac{-3\pi}{4}$

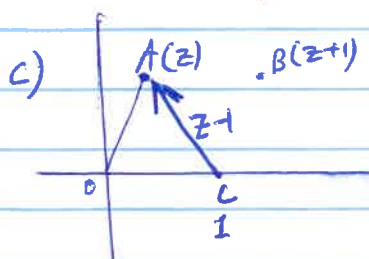
$\therefore -1 - i = \sqrt{2} \operatorname{cis}\left(\frac{-3\pi}{4}\right)$

ii) $(\sqrt{2})^{10} \operatorname{cis}\left(\frac{-30\pi}{4}\right)$

$= 32 \operatorname{cis}\left(\frac{2\pi}{4}\right)$

$= 32(\cos \frac{\pi}{2} + i \sin \frac{\pi}{2})$

$\therefore \text{real part} = 0.$



i) $|z| = 1, \therefore$ all sides of parallelogram are equal to 1.

$\therefore$ Rhombus.

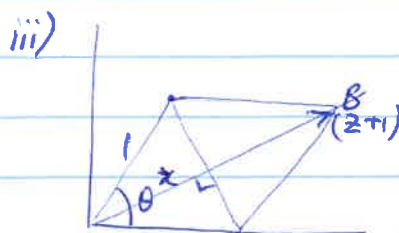
ii) see diagram.

$\arg\left(\frac{z-1}{z+1}\right) = \arg(z-1) - \arg(z+1)$

$= 90^\circ \left(\frac{\pi}{2}\right)$ Since

diagonals of rhombus are $\perp$.

$\therefore \frac{z-1}{z+1} = ki, k \in \mathbb{R}$ is purely imaginary.



$\arg(z+1) = \frac{\theta}{2}$ (diagonals of rhombus bisect $\angle$ it passes)

$x = \cos \frac{\theta}{2}$

$\therefore |z+1| = 2 \cos \frac{\theta}{2}.$

d) i) $z^7 - 1 = 0$

$(z-1)(z^6 + z^5 + z^4 + z^3 + z^2 + z + 1) = 0$

Since w is non real, $w \neq 1$

$\therefore w^6 + w^5 + w^4 + w^3 + w^2 + w + 1 = 0.$

ii) $w^7 + w^6 + w^5 + w^4 + w^3 + w^2 + w + 1 = 0$

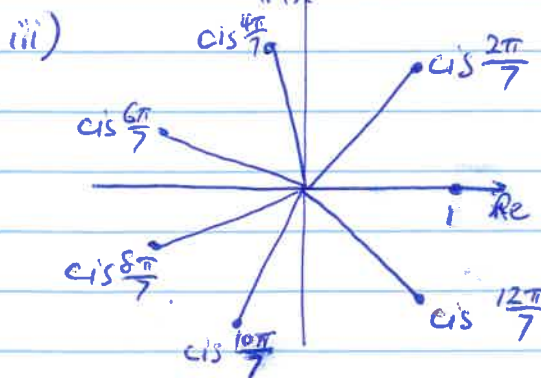
$= 1 + w^6 + w^4 + w + 1 + w^5 + w^3 + w^2 + 1$

$= 3 + (w^6 + w^5 + w^4 + w^3 + w^2 + w)$

$= 3 - 1$

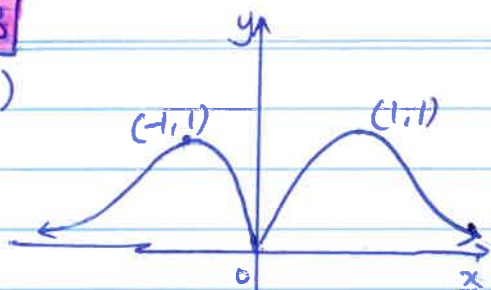
$= 2.$

Since $w^7 = 1$



Q13

a) i)



c) i) $\delta V = 2\pi r h \delta x$

$r = 2 - x$

$h = 2y$

$= 2\sqrt{1-x^2}$

$\delta V = 2\pi(2-x) \cdot 2\sqrt{1-x^2} \delta x$

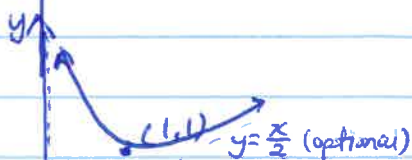
from $x^2 + y^2 = 1$

$= 4\pi(2-x)\sqrt{1-x^2} \delta x$

$V = \lim_{\delta x \rightarrow 0} \sum_{x=0}^1 \delta V$

$= 4\pi \int_0^1 (2-x)\sqrt{1-x^2} dx$ as req.

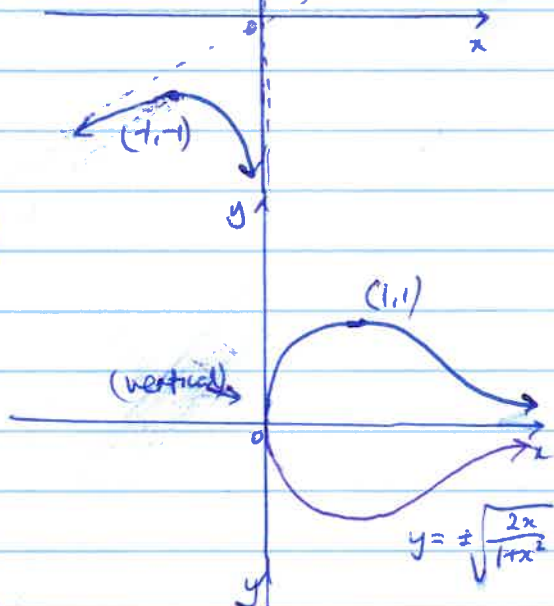
ii)



ii) $V = 8\pi \int_0^1 \sqrt{1-x^2} dx - 4\pi \int_0^1 x\sqrt{1-x^2} dx$

$= 8\pi \cdot \frac{\pi \times 1^2}{4} + 2\pi \int_0^1 -2x\sqrt{1-x^2} dx$

iii)



$= 2\pi^2 + 2\pi \int_1^0 \sqrt{u} du$

$u = 1 - x^2$
 $du = -2x dx$

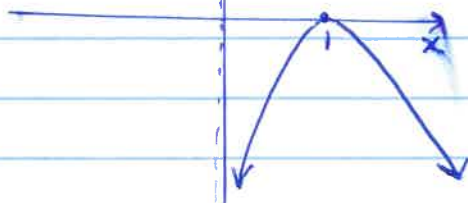
$x = 0, u = 1$
 $x = 1, u = 0$

$= 2\pi^2 + 2\pi \left[\frac{u^{3/2}}{3/2} \right]_1^0$

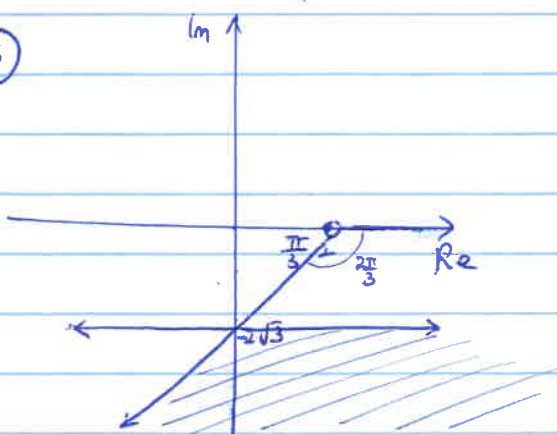
$= 2\pi^2 + 2\pi \left(0 - \frac{2}{3} \right)$

$= 2\pi^2 - \frac{4\pi}{3} \text{ units}^3$

iv)



b)



Q14

a) RTP $\frac{1}{x} + \frac{1}{y} \geq \frac{4}{x+y}$

$$\frac{y+x}{xy} \geq \frac{4}{x+y}$$

$$(x+y)^2 \geq 4xy$$

$$x^2 + 2xy + y^2 \geq 4xy$$

$$(x-y)^2 \geq 0$$

$$x^2 - 2xy + y^2 \geq 0$$

$$+4xy \quad +4xy$$

$$x^2 + 2xy + y^2 \geq 4xy$$

$$(x+y)^2 \geq 4xy$$

$$\frac{x+y}{xy} \geq \frac{4}{x+y} \quad (\text{since } x, y > 0)$$

$$\frac{1}{x} + \frac{1}{y} \geq \frac{4}{x+y} \quad \text{as req.}$$

b) iii) a. $0 = \frac{50-v}{5}$

$V \rightarrow 50$ m/s is terminal velocity.

iv) $v \frac{dv}{dx} = \frac{50-v}{5}$

$$\frac{dv}{dx} = \frac{50-v}{5v}$$

$$\frac{dx}{dv} = \frac{5v}{50-v}$$

$$x = \int \frac{5v}{50-v} dv$$

$$= -5 \int \frac{-v+50-50}{50-v} dv$$

$$= -5 \int 1 dv + 250 \int \frac{-1}{50-v} dv$$

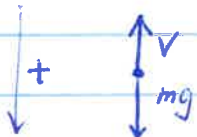
$$x = -5v - 250 \ln(50-v) + c$$

$$x=0, v=0 : c = 250 \ln 50$$

$$x = -5v - 250 \ln(50-v) + 250 \ln 50$$

$$= -5v + 250 \left(\frac{50}{50-v} \right) \quad \text{as req.}$$

b) i)



$$F_{\text{net}} = mg - v$$

$$ma = mg - v$$

$$5a = 50 - v$$

$$a = \frac{50-v}{5}$$

ii) $\frac{dv}{dt} = \frac{50-v}{5}$

$$\frac{dt}{dv} = \frac{5}{50-v}$$

$$t = 5 \int \frac{-1}{50-v} dv$$

$$t = -5 \ln(50-v) + c$$

$$t=0, v=0$$

$$0 = -5 \ln 50 + c$$

$$c = 5 \ln 50$$

$$t = -5 \ln(50-v) + 5 \ln 50$$

$$t = -5 \ln \left(\frac{50-v}{50} \right)$$

$$\frac{50-v}{50} = e^{-\frac{t}{5}}$$

$$50-v = 50e^{-\frac{t}{5}}$$

$$v = 50 - 50e^{-\frac{t}{5}}$$

c) i) $\angle KMQ = \angle KAB = \theta$ (exterior $\angle$ of cyclic quad is equal to interior opposite $\angle$)

ii) $\angle KPO = \angle KMO = \theta$ ($\angle$ s standing on same arc are equal)

$\therefore \angle KPO = 180 - \theta$ ($\angle$ s on a straight line)

$\therefore AKPO$ is cyclic (opposite $\angle$ s add to 180°)

iii) $\angle ABM = 180 - \phi$ (opposite $\angle$ s of cyclic quad add to 180°)

$\angle MPQ = 180 - \phi$ (exterior $\angle$ of cyclic quad is equal to interior opposite $\angle$)
(BMPO)

$\therefore \angle QKM = 180 - \phi$ (---) (KMPO)

$\therefore \angle QKA = 180^\circ$ (from $\angle AKM + \angle QKM$)

$\therefore AKQ$ is collinear if BMPO is cyclic.

Q15

a) i) The triangle inequality:

$$a+b > c \Rightarrow a > c-b$$

(only if $c \geq b$): $a^2 > (c-b)^2$

$$a+c > b \Rightarrow a > b-c$$

(only if $b \geq c$): $a^2 > (b-c)^2$

In either case $a^2 > (b-c)^2$

(Arguments using $|b-c|$ accepted)

(if diagram or cases shown)

ii) $a^2 > (b-c)^2$

$$a^2 > b^2 - 2bc + c^2$$

Similarly $b^2 > a^2 - 2ac + c^2$

$$c^2 > b^2 - 2ab + a^2$$

$$a^2 + b^2 + c^2 > 2a^2 + 2b^2 + 2c^2 - 2(ab + ac + bc)$$

$$2ab + 2ac + 2bc > a^2 + b^2 + c^2$$

$$(+2ab + 2ac + 2bc) \quad (+2ab + 2ac + 2bc)$$

$$4(ab + bc + ca) > (a + b + c)^2 \text{ as req.}$$

b) i) $x^2 - y^2 = 1$, $a=1, b \neq 1$

$$b^2 = a^2(e^2 - 1)$$

$$e^2 = 2$$

$$e = \sqrt{2}$$

$$\frac{dy}{dx} = \frac{dy/d\theta}{dx/d\theta} \quad P(\sec\theta, \tan\theta)$$

$$= \frac{\sec^2\theta}{\sec\theta \tan\theta}$$

$$= \frac{\sec\theta}{\tan\theta}$$

$$= \frac{1}{\sin\theta}$$

$$m_{\perp} = -\sin\theta$$

eqn normal: $y - \tan\theta = -\sin\theta(x - \sec\theta)$

$$y - \tan\theta = -x\sin\theta + \tan\theta$$

$$y = -x\sin\theta + 2\tan\theta$$

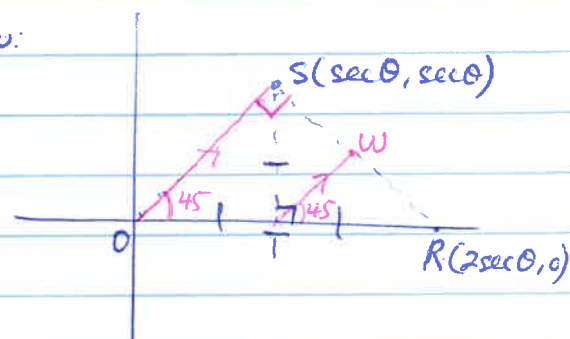
as req.

b) ii) Eqn OS: $y=x$

$\therefore$ at S, x-coord is $\sec\theta$

$$\therefore S(\sec\theta, \sec\theta)$$

Now:



$$R: y=0: 0 = -x\sin\theta + 2\tan\theta$$

$$x\sin\theta = 2\tan\theta$$

$$x = \frac{2}{\cos\theta} = 2\sec\theta$$

$\therefore$ On diagram above $\triangle SRT$ is right

Isosceles: $\therefore \angle SRT = 45^\circ$

$$\therefore \frac{TR}{SR} = \sin 45^\circ$$

$$\therefore \sqrt{2} \cdot TR = RS$$

ii) On diagram: $OT = TR \Rightarrow SW = WR$

(Intercepts on transversals cut by parallel lines in same ratio)

$$\therefore W \text{ is } M_{SR}: \left(\frac{\sec\theta + 2\sec\theta}{2}, \frac{\sec\theta}{2} \right)$$

$$W: \left(\frac{3}{2}\sec\theta, \frac{1}{2}\sec\theta \right)$$

iv) TW is already perp. bisector of SR.

$\therefore F$ must be T:

$$\therefore ae = \sec\theta$$

$$\sqrt{2} = \sec\theta$$

$$\theta = 45^\circ \text{ or } 315^\circ$$

ie

$$\frac{\pi}{4} \text{ or } \frac{7\pi}{4}$$

Q15c)

c) i) Area segment := $\frac{1}{2}r^2(\theta - \sin\theta)$

Area triangle = $\frac{1}{2}r^2\sin\theta$

$A = \frac{1}{2}r^2\sin\theta - \frac{1}{2}r^2(\theta - \sin\theta)$

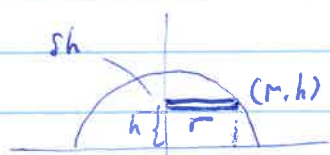
$= r^2\sin\theta - \frac{1}{2}r^2\theta$

$= r^2(\sin\theta - \frac{\theta}{2})$

ii) $\theta = \frac{\pi}{3}$

$\therefore A = r^2(\sin\frac{\pi}{3} - \frac{\pi}{6}) \times 6$

$= r^2(\frac{\sqrt{3}}{2} - \frac{\pi}{6}) \times 6$



$r^2 + h^2 = 4$

$r = \sqrt{4 - h^2}$

$A = (\sqrt{4 - h^2})(\frac{\sqrt{3}}{2} - \frac{\pi}{6}) \times 6$

$= (\frac{\sqrt{3}}{2} - \frac{\pi}{6})(4 - h^2) \times 6$

$V = \lim_{h \rightarrow 0} \sum_{n=0}^{\infty} A \delta h$

$= 6 \int_0^2 (\frac{\sqrt{3}}{2} - \frac{\pi}{6})(4 - h^2) dh$

$= 6(\frac{\sqrt{3}}{2} - \frac{\pi}{6}) [4h - \frac{h^3}{3}]_0^2$

$= 6(\frac{\sqrt{3}}{2} - \frac{\pi}{6})(8 - \frac{8}{3})$

$= (\frac{\sqrt{3}}{2} - \frac{\pi}{6})(32)$

$= 16\sqrt{3} - \frac{16\pi}{3} \text{ units}^3$

Q16

a) $I_n = \int_0^1 x^{2n} x e^{-x^2} dx$

i)

$u = x^{2n}$

$\therefore v = \frac{1}{2}e^{-x^2}$

$u' = 2nx^{2n-1}$

$v' = -x e^{-x^2}$

$v = \int -x e^{-x^2} dx$

$= -\frac{1}{2}e^{-x^2} x$

$I_n = [x^{2n} \times \frac{1}{2}e^{-x^2}]_0^1 - \int_0^1 \frac{1}{2}e^{-x^2} \cdot 2nx^{2n-1} dx$

$= \frac{1}{2e} + n \int_0^1 e^{-x^2} x^{2n-1} dx$

$= \frac{1}{2e} + n \int_0^1 e^{-x^2} x^{2(n-1)+1} dx$

$= \frac{1}{2e} + n I_{n-1} \quad \checkmark \text{ as req.}$

ii) $I_0 = \int_0^1 x e^{-x^2} dx$

$= [-\frac{1}{2}e^{-x^2}]_0^1$

$= -\frac{1}{2}e^{-1} - (-\frac{1}{2}e^0)$

$= \frac{1}{2e} + \frac{1}{2} \text{ as req.}$

iii) Let $n=1$

LHS = $1 + \frac{1}{1!}$

$= 2$

RHS = $e - \frac{2eI_1}{1!}$

$= e - 2eI_1$

$I_1 = \frac{1}{2e} + 1I_0$

$= \frac{1}{2e} + \frac{1}{2e} + \frac{1}{2}$

$= \frac{1}{e} + \frac{1}{2}$

RHS = $e - 2e(\frac{1}{e} + \frac{1}{2})$

$= e - e + 2$

$= 2$

16a) iii) cont

Assume for $n=k$:

$$1 + \frac{1}{1!} + \dots + \frac{1}{k!} = e - \frac{2eI_k}{k!}$$

RTP: for $n=k+1$

$$1 + \frac{1}{1!} + \dots + \frac{1}{k!} + \frac{1}{(k+1)!} = e - \frac{2eI_{k+1}}{(k+1)!}$$

$$\text{LHS} = e - \frac{2eI_k}{k!} + \frac{1}{(k+1)!}$$

$$= e - \frac{2eI_k(k+1) + 1}{(k+1)!}$$

Now from i): $I_{k+1} = -\frac{1}{2e} + (k+1)I_k$

$$\therefore 2eI_{k+1} = -1 + 2e(k+1)I_k$$

$$= e - \frac{2eI_{k+1}}{(k+1)!} \text{ as required.}$$

true for $n \geq 1$ by mathematical induction.

iv) For $0 \leq x \leq 1, 0 \leq x^{2n+1} \leq 1$

and $0 \leq e^{-x^2} \leq 1$

$$I_n = \int_0^1 x^{2n+1} e^{-x^2} dx \leq \int_0^1 1 dx$$

(upper bound)

$$0 \leq I_n \leq 1 \text{ for } n \geq 0$$

As $n \rightarrow \infty$

$$\frac{2eI_n}{n!} \text{ vanishes.}$$

$$\therefore e - \frac{2eI_n}{n!} \rightarrow e \text{ as } n \rightarrow \infty$$

16b) i) $xy + xz + zy = \frac{(x+y+z)^2 - (x^2+y^2+z^2)}{2}$

$$= \frac{5^2 - 8}{2}$$
$$= \frac{17}{2}$$

ii) consider: $(xy + xz + zy)(x+y+z)$

$$(x+y+z)(x^2+y^2+z^2) = 5 \times 8$$

$$x^3 + xy^2 + xz^2 + y^3 + yx^2 + yz^2 + z^3 + zx^2 + zy^2 = 40$$

$$\therefore xy^2 + xz^2 + yz^2 + yx^2 + zx^2 + zy^2 = 40 - (x^3 + y^3 + z^3)$$

$$= 40 - 13$$

$$= 27$$

Q166 cont)

iii) Consider:

$$(x+y+z)(xy+yz+zx)$$

$$= x^2y + x^2z + xyz + xy^2 + xyz + y^2z + xyz + xz^2 + yz^2$$

$$= 27 + 3xyz \quad (ii)$$

$$\therefore 5\left(\frac{17}{2}\right) = 27 + 3xyz$$

↑
(i)

$$\therefore \frac{85}{2} - 27 = 3xyz$$

$$\frac{31}{2} = 3xyz$$

$$xyz = \frac{31}{6}$$

iv) To find $x^4+y^4+z^4$ first consider:

$$\frac{(x+y+z)(x^3+y^3+z^3)}{5 \times 13} = \frac{x^4+y^4+z^4}{X} + \underbrace{xyz(x^2+y^2+z^2)}_{\textcircled{1} Y}$$

To find $\textcircled{1}$: consider:

$$\begin{aligned} \frac{(xy+xz+yz)(x^2+y^2+z^2)}{\frac{17}{2} \times 8} &= x^3y + xy^3 + xyz^2 + x^3z + xzy^3 + xz^3 \\ &\quad + yzx^2 + y^3z + yz^3 \\ &= x^3y + xy^3 + x^3z + xz^3 + y^3z + yz^3 \\ &\quad \textcircled{1} Y + xyz(x+y+z) \end{aligned}$$

$$\therefore \frac{17}{2} \times 8 = Y + \frac{31}{6} \times 5$$

$$\begin{aligned} Y &= 68 - \frac{155}{6} \\ &= \frac{253}{6} \end{aligned}$$

$$\begin{aligned} \therefore X &= 65 - \frac{253}{6} \\ &= \frac{137}{6} \end{aligned}$$